Project Design Phase-II

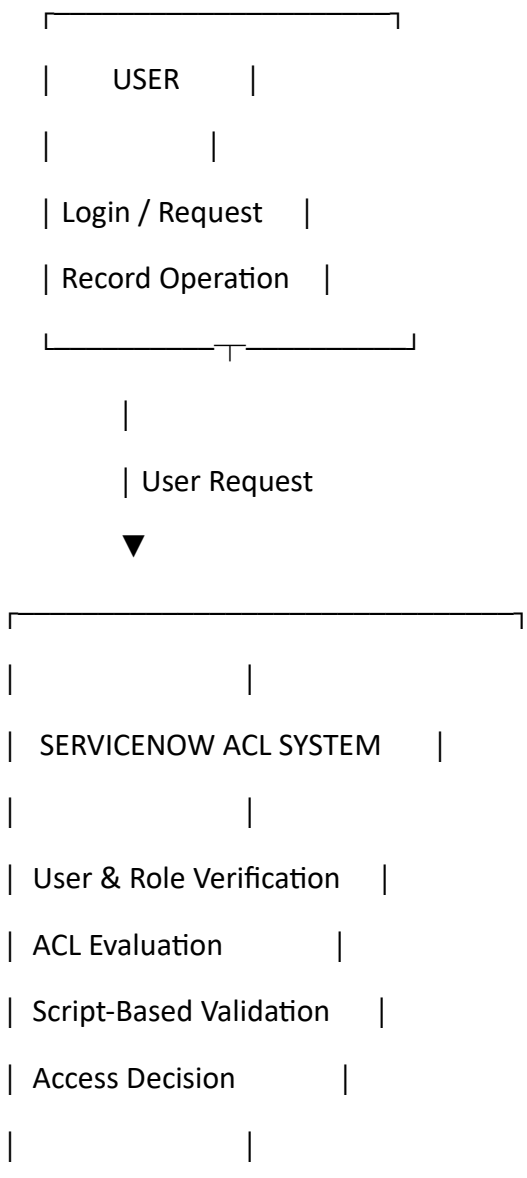
Data Flow Diagram & User Stories

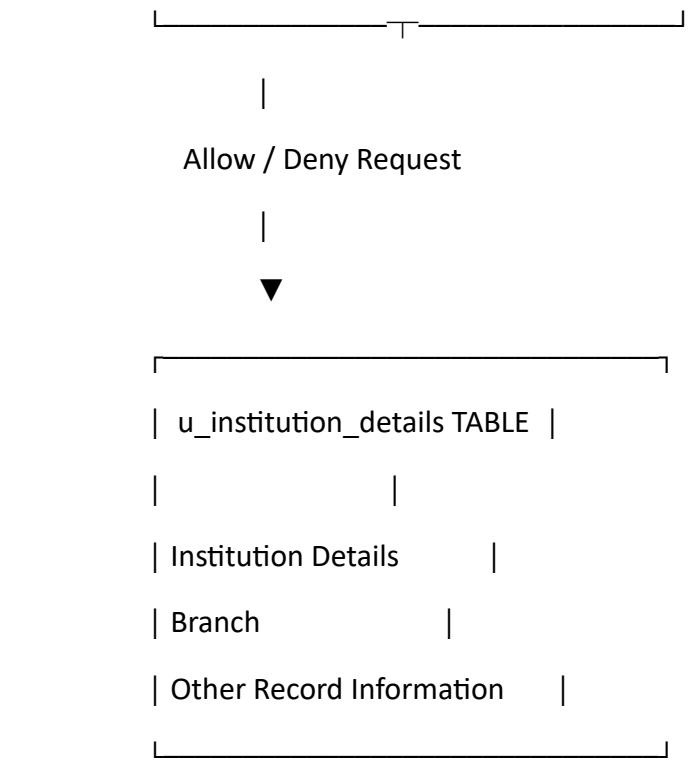
Date: 03 October 2026
Team ID: 6aba117e538da4003d63184c
Project Name: Script-Controlled ACL – Restrict Record Access Based on Field Value
Maximum Marks: 4 Marks

Data Flow Diagram (DFD)

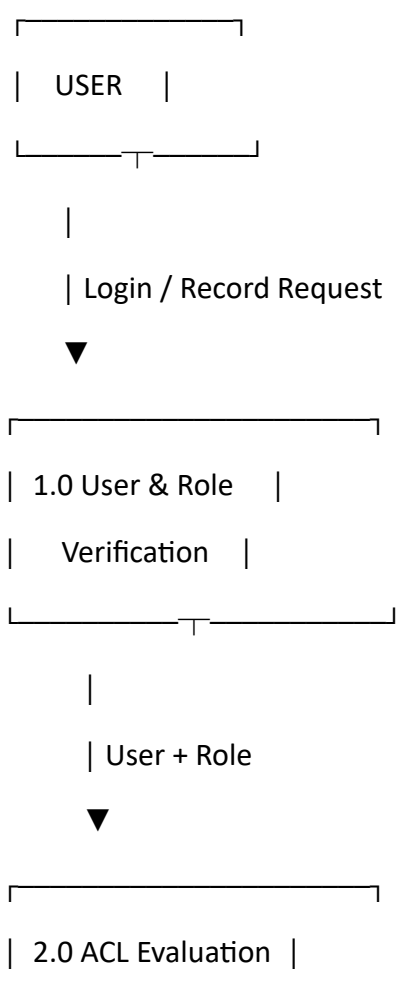
A Data Flow Diagram represents how users, roles, ACL rules, scripts, and records interact within the proposed ServiceNow access-control system.

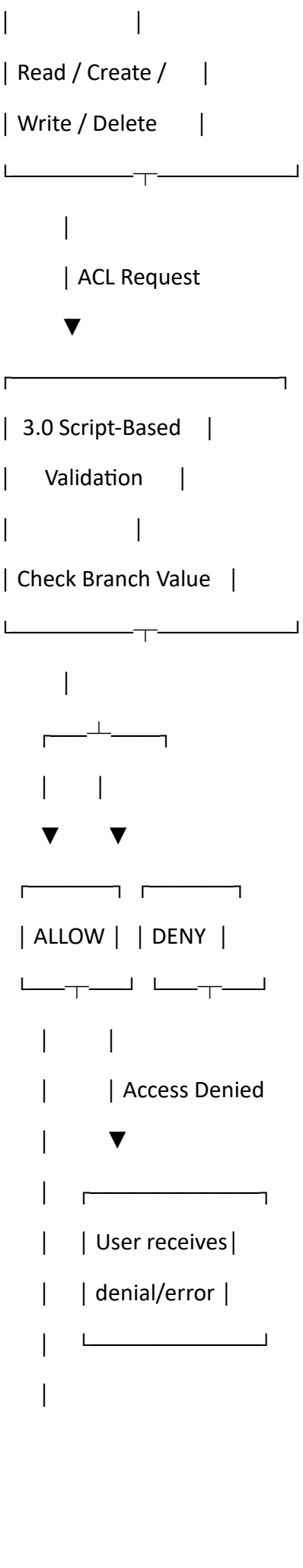
Level-0 DFD / Context Diagram

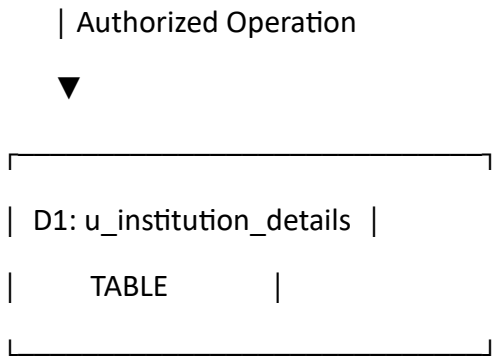



Allow / Deny Request
u_institution_details TABLE
Institution Details
Branch
Other Record Information

Level-1 DFD


USER
Login / Record Request
1.0 User & Role Verification
User + Role
2.0 ACL Evaluation


Read / Create /
Write / Delete
ACL Request
3.0 Script-Based
Validation
Check Branch Value
ALLOW
DENY
Access Denied
User receives
denial/error



DFD Data Stores and Processes

Element	Description
External Entity	User
Process 1.0	User & Role Verification
Process 2.0	ACL Evaluation
Process 3.0	Script-Based Branch Validation
Data Store D1	u_institution_details table
Output 1	Authorized access
Output 2	Access denied

Note: Since this project is implemented within ServiceNow, the user/role management, ACL engine, scripts and database are all components of the ServiceNow platform. There is no need to show unrelated external APIs or third-party systems.

User Stories

Use the following table in your report.

User Type	Functional Requirement (Epic)	User Story Number	User Story / Task	Acceptance Criteria	Priority	Release
User	User & Role Management	**USN-1**	As a user, I want to log in to ServiceNow so that my assigned roles and	User can successfully log in and ServiceNow identifies	High	Sprint-1

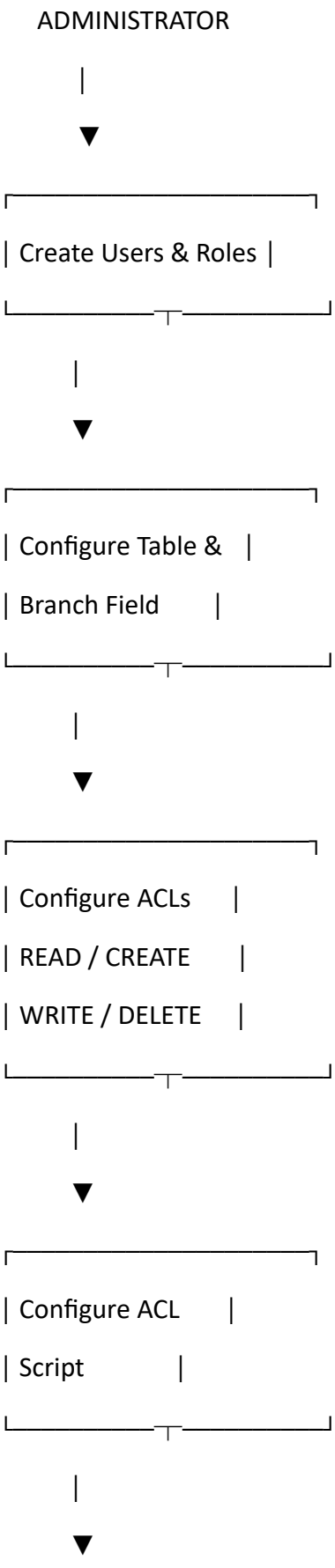
			permissions can be identified.	the assigned roles.		
Administrator	User & Role Management	**USN-2**	As an administrator, I want to create and manage users so that appropriate users can access the system.	Administrator can create users and maintain their account details.	High	Sprint-1
Administrator	User & Role Management	**USN-3**	As an administrator, I want to assign appropriate roles to users so that access can be controlled based on their responsibilities.	User receives the assigned role and corresponding permissions.	High	Sprint-1
Administrator	Table & Record Management	**USN-4**	As an administrator, I want to create the u_institution_details table so that institution information can be stored and managed.	Table is created with the required fields and can store records.	High	Sprint-1
Administrator	Table & Record Management	**USN-5**	As an administrator, I want to add the Branch field to institution records so that access can be controlled based on branch values.	Branch field is available and records contain valid branch information.	High	Sprint-1

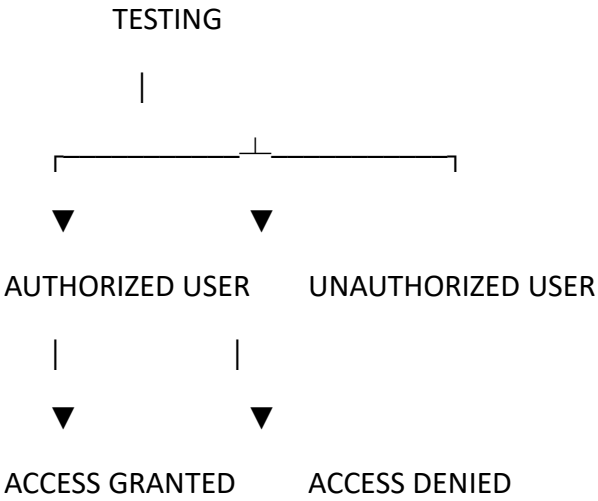
User	READ Access Control	**USN-6**	As a user, I want the system to verify my permission before viewing institution records so that unauthorized information cannot be accessed.	Authorized users can view permitted records; unauthorized users are denied access.	High	Sprint-2
User	CREATE Access Control	**USN-7**	As a user, I want the system to verify my permission before creating a record so that only authorized users can add institution details.	Authorized users can create records; unauthorized users cannot create records.	High	Sprint-2
User	WRITE Access Control	**USN-8**	As a user, I want the system to verify my permission before modifying a record so that unauthorized changes are prevented.	Authorized users can update permitted records; unauthorized users cannot modify them.	High	Sprint-2
User	DELETE Access Control	**USN-9**	As a user, I want the system to verify my permission before deleting a record so that important data cannot be deleted without authorization.	Authorized users can delete permitted records; unauthorized users cannot	High	Sprint-2

				delete records.		
Administrator	Script-Based Access Restriction	**USN-10**	As an administrator, I want to use an ACL script to validate the Branch field so that access can be restricted according to branch-specific rules.	ACL script checks the Branch value and allows or denies access according to the defined rule.	High	Sprint-2
Administrator	Script-Based Access Restriction	**USN-11**	As an administrator, I want restricted branch records to be inaccessible to unauthorized users so that sensitive records remain protected.	Users without the required authorization cannot access restricted branch records.	High	Sprint-2
Administrator	Testing & Validation	**USN-12**	As an administrator, I want to test all ACL operations so that the access-control implementation works as expected.	READ, CREATE, WRITE and DELETE operations are tested with authorized and unauthorized users and produce the expected results.	High	Sprint-3

Simplified User Story Flow

You can also show this below the table:




TESTING
AUTHORIZED USER
UNAUTHORIZED USER
ACCESS GRANTED
ACCESS DENIED